EXP 5: Implementation of K-Nearest Neighbours (K-NN)

CODE

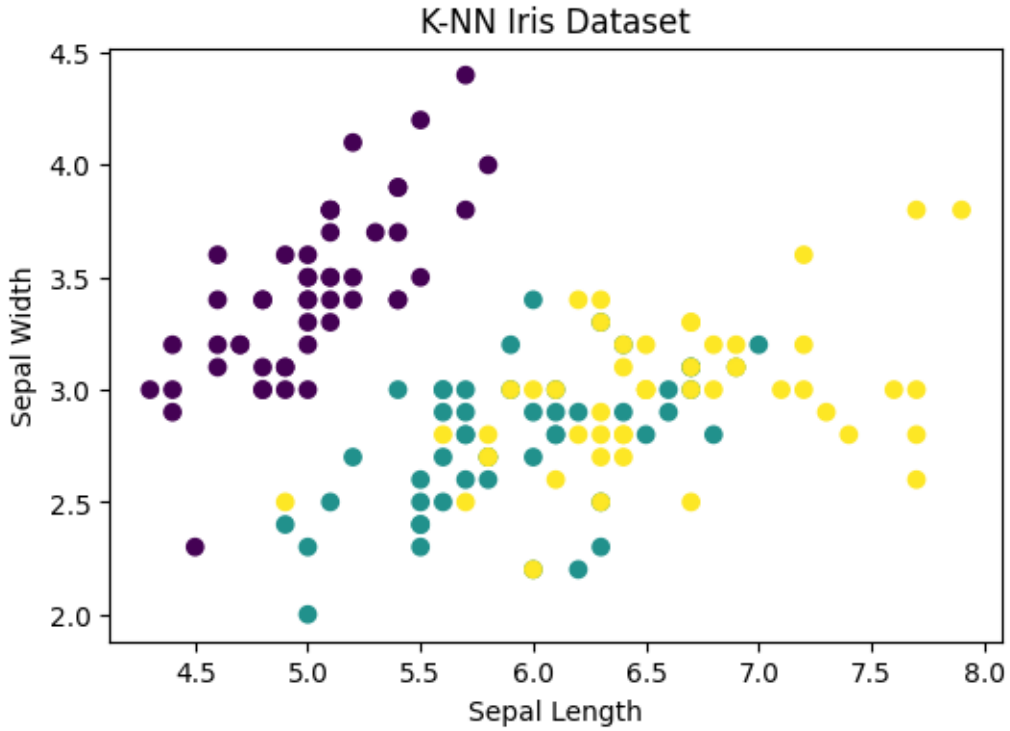
```
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
iris = load_iris()
X = iris.data[:, :2]  # Use two features for graph
y = iris.target
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
model = KNeighborsClassifier(n_neighbors=3)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("K-NN Classification")
print("K Value: 3")
print("Accuracy:", round(accuracy_score(y_test, y_pred) * 100, 2), "%")
plt.figure(figsize=(6, 4))
plt.scatter(X[:, 0], X[:, 1], c=y, cmap='viridis')
plt.xlabel("Sepal Length")
plt.ylabel("Sepal Width")
plt.title("K-NN Iris Dataset")
plt.show()
sample = [[5.1, 3.5]]
prediction = model.predict(sample)
print("New Sample Prediction:", iris.target_names[prediction[0]])
```

OUTPUT

K-NN Classification

K Value: 3

Accuracy: 83.33 %



New Sample Prediction: setosa